BINDR

Biomolecular Interactions & Drug Research

- **The Mission:** Transform 4 to 8 weeks of fragmented biopharma investment diligence into an adversarial 60-second triage.
- **The Core Thesis:** 90% of drugs fail for three independent reasons: false biological target, trial execution failure, or post-launch resistance mutations. Bindr surfaces which failure mode bites before capital is committed.
- **Architecture:** 3 parallel specialist sub-agents · Deterministic Python mathematical verification · Gemini 4 Argon Interactions API · Human Analyst Sign-Off Gate · 1-Page Diligence Memo export.
- **Validation:** Evaluated on historical cohorts with strict *as_of* temporal cutoffs. All 3 checks beat published/random baselines (Science AUROC 0.790, p=0.028; Trial Precision@20 0.400; Resistance Recall@10 0.500).
- **Live Web Application:** https://bindr-coc2.onrender.com · 127/127 automated tests passing on main.

The Problem: The Biopharma Diligence Bottleneck

90% of clinical-stage drug assets fail. Traditional diligence is manual, slow, and backward-looking.

The Diligence Bottleneck

• Pharma BD and Biotech VC teams evaluate 100+ pre-clinical and Phase 1/2 assets each year.
• Manual diligence takes **4 to 8 weeks** per asset across fragmented academic literature, trial registries, and structure databases.
• A single Phase 2/3 trial failure destroys **0M–00M+** in capital and years of development timeline.

The 3 Independent Failure Modes

1. **Target & Biology Risk (Lab Stage):** Target is not causally linked to disease in humans (Phase 2 lack of efficacy).
2. **Trial Execution Risk (Trial Stage):** Study terminates early due to patient accrual bottlenecks, futility, or narrow therapeutic window.
3. **Acquired Resistance Risk (Market Stage):** Tumors mutate in patients post-launch, sterically blocking drug binding.

Why AI Tools Fail Today

• **LLM Hallucinations:** Generic LLMs invent clinical trial statistics and binding affinities.
• **Hindsight Bias (Data Leakage):** Models evaluate past drugs using data published years after trial start.
• **Lack of Auditability:** Black-box models cannot cite immutable primary database records.

The Solution: Bindr's 3 Specialist Sub-Agents

An adversarial multi-agent triage pipeline executing in parallel in <60 seconds.

Sub-Agent 1: Target & Biology

Open Targets · ChEMBL · DepMap

- Evaluates human genetic evidence score, somatic mutations, and constraint (LOEUF).
- Quantifies cancer dependency across 1,000+ DepMap cell lines.
- Audits target safety liabilities to flag on-target organ toxicity.

Sub-Agent 2: Trial Execution

ClinicalTrials.gov Protocol Registry

- Queries both direct asset studies and disease indication-level attrition benchmarks.
- NLP regex classifier categorizes *whyStopped* rationales (Accrual, Futility, Safety, Commercial, Regulatory).
- Identifies patient recruitment bottlenecks before capital commitment.

Sub-Agent 3: 3D Resistance

RCSB PDB · AlphaFold DB · UniProtKB

- Downloads experimental 3D X-ray co-crystals or AlphaFold predictions via gemmi.
- Python contact-distance engine computes Euclidean heavy-atom distances (Å) between ligand and residues.
- Maps clinical missense variants to identify active-site escape routes.

System Architecture & Google DeepMind Gemini Integration

Built strictly on the Gemini Interactions API with deterministic mathematical verification.

Interactions API with Multi-Turn State

• Uses google-genai >= 2.3.0 Interactions API passing previous_interaction_id.
• **gemini-4-argon:** Deep multi-turn reasoning over the Evidence Ledger for diligence synthesis.
• **gemini-3.8-flash:** Rapid deterministic entity resolution and sub-agent routing.

Thinking Configuration Standards

• Reasoning depth configured via generation_config={"thinking_level": "high"}.
• Deprecated parameters (temperature, top_p, top_k, thinking_budget) are completely forbidden by architectural unit tests.

Deterministic Python Math Gate

• **LLMs never assert numbers.**
• Combined risk probability P(failure) = 1 - ∏(1 - s_i), 3D contact distances (Å), and statistical metrics are strictly computed in Python.

Evidence Ledger & Sign-Off Lock

• Every claim is an atomic LedgerRow citing an external URL, date, and licence.
• Model prose is verified against ledger IDs. Final diligence reports require a **Human Analyst Sign-Off Gate** before export.

Live Interactive Platform & 3D Resistance Viewer

Production single-command web application live at https://bindr-coc2.onrender.com

1. Hands-Free Voice Query

• Browser Web Speech API integration.
• Analysts speak drug, target, and disease naturally; instantly parsed into canonical identifiers.

2. 3D Protein Map (3Dmol.js)

• Real-time 3D protein cartoon ribbon with cyan ligand sticks.
• Color-coded escape mutations: ≤2.5 Å direct contact (critical red), 2.5–5 Å pocket (high risk), 5–8 Å rim (moderate).
• Clicking mutation table flies 3D camera to zoom into that residue.

3. Human Analyst Sign-Off Gate

• Named reviewer reviews each AI red flag.
• Accept / Reject buttons (with mandatory audit trail rationale).
• Unlocks executive 1-Page Diligence Memorandum (Markdown copy & PDF print).

Tested Against a Baseline, With No Hindsight (Empirical Proof)

Every check evaluated on real historical cohorts with strict as_of temporal cutoffs.

Check Lane	Cohort / Ground Truth	Published Baseline	Bindr Empirical Result	Key Proof Point
Target Biology (Science)	20 historical programmes (10 failed Phase 2/3 vs. 10 approved)	AUROC 0.500 (Phase 2->3 base rate)	**AUROC 0.790 (Separation: +0.127)**	Statistically significant separation (Mann-Whitney p=0.028, n=20) under strict historical as_of cutoffs.
Trial Execution (Trial)	1,350 post-2022 test trials vs. 2,000 pre-2022 graveyard	Precision@20: 0.370 (Cohort prevalence)	**Precision@20: 0.400**	Beats prevalence baseline on top-20 riskiest trials. Out-of-period AUROC 0.479 reported as honest negative result.
3D Resistance (Market)	7 approved kinase inhibitors, 14 clinical escape mutations	Recall@10: 0.158 (pocket) 0.041 (whole chain)	**Recall@10: 0.500 (7/14 escapes found)**	**3.17x enrichment** over pocket baseline. Beats random baselines at every evaluated k from 5 to 20.
Unseen Stress Test	10 diverse unseen assets (oncology, rare, metabolic)	—	**10 / 10 Valid Verdicts (3.69s wall clock)**	Median latency 0.66s. Demonstrates honest degradation (biologics/peptides degrade cleanly to Not Scored).

Voluntary Failure Case & Data Integrity

Transparent disclosure of scientific boundaries and zero-data-leakage verification.

Where We Get It Wrong: Imatinib M351T

- **The Failure:** Imatinib resistance via residue **M351T** sits at **8.5 Å** from the drug in ABL1 coordinates.
- **Why We Missed It:** M351T causes resistance through an **allosteric conformational shift** rather than direct contact.
- **The Lesson:** Pure Euclidean contact distance methods miss allosteric resistance. Rather than hiding this failure, Bindr documents it openly.

Zero Date Leakage Enforcement

- Every historical asset is scored under a strict as_of cutoff date set before clinical outcomes were known.
- EvidenceLedger(as_of=...) automatically rejects any row dated after the cutoff.
- Zero temporal violations confirmed across all backtest runs.

7 Public Dataset Licences Audited

- **Open Targets Platform:** Creative Commons CC0 1.0
- **ChEMBL Database:** Creative Commons CC BY-SA 3.0
- **ClinicalTrials.gov:** U.S. Public Domain
- **RCSB Protein Data Bank:** Creative Commons CC0 1.0
- **UniProtKB / Swiss-Prot:** Creative Commons CC BY 4.0
- **DepMap CRISPR Dependency:** Creative Commons CC BY 4.0

Market Opportunity & Commercial Business Model

Targeting the high-margin biopharma intelligence and diligence sector.

Target Customers & Market Size

• **3,500+ Institutions:** Biotech Venture Capital, Pharma Corporate BD teams, and Life Science Hedge Funds.
• **TAM:** .2B global biopharma diligence and competitive intelligence market.

High ROI Value Proposition

• Compresses 4–8 weeks of manual diligence into **60 seconds**.
• Prevents **0M+** Phase 2/3 trial failures by highlighting hidden recruitment bottlenecks and active-site mutational escape routes early.

SaaS & Enterprise Pricing Model

• **Tier 1 (SaaS Subscription):** ,500 / analyst seat / month (unlimited triage runs, 3D structure maps, exportable memos).
• **Tier 2 (Enterprise API):** Custom integrations into internal proprietary pipeline databases and compound screening libraries.

Team, Live Demo & Conclusion

Bindr — Know why it fails before you commit capital.

Zoravar Singh (Zoro) Lead Integration & Core Architecture Orchestrator, Evidence Ledger, Science Check, Gemini 4 Argon wiring	**Ayman** Clinical Trial Check ClinicalTrials.gov API v2, graveyard benchmarking, whyStopped NLP classifier
Rishi Market & Resistance Check RCSB PDB & AlphaFold 3D Euclidean contact distances, UniProt variant mapping	**Krystal** Frontend Architecture Single-page web application, 3Dmol.js protein viewer, analyst sign-off UX
Khaleem Backtest Evaluation & Licencing Statistical methodology, historical cohorts, 10-asset stress test, evaluation report	**Google DeepMind Track Checklist** ✓ Gemini Interactions API (4-Argon & 3.8-Flash) ✓ thinking_level high · No deprecated parameters ✓ Python owns all numbers · Verified Evidence Ledger